SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

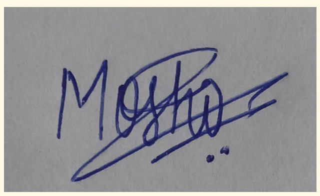
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	Moshika M	**Reg. No.:**	192424064
Section / Batch:	B.Tech (AI-DS)	**Date:**	07.08.2026

Code of Conduct

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*
The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ_1 = 0.5 (Trigram)
- λ_2 = 0.3 (Bigram)
- λ_3 = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

ANSWER:

1. **Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.**

Formula:
P(science | data) = C(data science) / C(data)

Given:
C(data science) = 3
C(data) = 3

P(science | data) = 3 / 3 = 1.0

Answer:
P(science | data) = 1.0 or 100%

Interpretation:

The word "science" follows "data" in all three occurrences in the corpus. Therefore, the keyboard can predict "science" with very high confidence whenever the user types "data". This improves prediction accuracy and reduces typing effort.

2. **A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.**

The trigram "data science improves" does not occur in the corpus. Therefore:

P(improves | data science) = 0

The model backs off to the bigram:

P(improves | science) = 0

Since "improves" also does not occur after "science", the model further backs off to the unigram probability:

P(improves) ≈ 0

Thus, the estimated probability of the sequence is very small or approximately zero without smoothing.

Backoff is essential because it allows the system to handle unseen word sequences. It prevents the language model from completely failing when users enter new or uncommon phrases. This is important for real-time keyboards because users continuously type new names, phrases, and expressions.

3. **Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.**

Formula:

P(is | data science) =
λ_1 P(is | data science) + λ_2 P(is | science) + λ_3 P(is)

Given:
$\lambda_1 = 0.5$
$\lambda_2 = 0.3$
$\lambda_3 = 0.2$

Trigram probability:

P(is | data science) = C(data science is) / C(data science)
= 2 / 3
= 0.667

Bigram probability:

P(is | science) = C(science is) / C(science)
= 2 / 3
= 0.667

Unigram probability:

Total number of words = 12
Number of occurrences of "is" = 2

P(is) = 2 / 12
= 0.167

Therefore,

P = (0.5 × 0.667) + (0.3 × 0.667) + (0.2 × 0.167)

P = 0.3335 + 0.2001 + 0.0334

P ≈ 0.567

Answer:

P(data science is) ≈ 0.567

Interpolation combines trigram, bigram, and unigram probabilities. The trigram provides more contextual information, while the lower-order models provide coverage when higher-order n-grams are sparse. This gives a balance between prediction accuracy and handling unseen or rare sequences.

4. **Calculate the Entropy of the distribution P(is) = 0.66 and P(drives) = 0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.**

Formula:

H = -Σ P(x) log2 P(x)

H = -[(0.66 log2 0.66) + (0.33 log2 0.33)]

H = -[(0.66 × -0.599) + (0.33 × -1.599)]

H = 0.395 + 0.528

H ≈ 0.923 bits

Answer:

Entropy ≈ 0.923 bits

The entropy value indicates the uncertainty in predicting the next word. A lower entropy means the model is more confident about its prediction, while a higher entropy indicates greater uncertainty. Entropy therefore helps evaluate the confidence and reliability of predictions made by an intelligent keyboard system.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.
Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.
Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

ANSWER:

1. **Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.**

Sentence 1: "Book a flight ticket now."

Book/VB
a/DT
flight/NN
ticket/NN
now/RB

Sentence 2: "This book is interesting."

This/DT
book/NN
is/VBZ
interesting/JJ

In the first sentence, "Book" is used as a command or action, so it is tagged as VB (Verb). In the second sentence, "book" refers to a physical object, so it is tagged as NN (Noun). The surrounding words provide the context needed to distinguish the two meanings.

2. **Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.**

Given:

$P(\text{book} \mid VB) = 0.6$
$P(\text{Start} \rightarrow VB) = 0.5$

Formula:

$P(VB, \text{book}) = P(\text{Start} \rightarrow VB) \times P(\text{book} \mid VB)$

$= 0.5 \times 0.6$

$= 0.30$

For comparison:

$P(NN, \text{book}) = P(\text{Start} \rightarrow NN) \times P(\text{book} \mid NN)$

$= 0.5 \times 0.4$

$= 0.20$

Since $0.30 > 0.20$, the HMM favors the VB tag.

Answer:

Likelihood of "book" as VB = 0.30

The model favors VB because the probability of observing "book" given the Verb tag is higher than the probability given the Noun tag. The sentence context also supports its use as a command.

3. **Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.**

Rule-Based Tagging uses manually defined grammatical rules to assign POS tags. It is simple and easy to understand but requires many rules and may not handle complex language variations effectively.

Stochastic or HMM Tagging uses probabilities learned from training data. It can handle ambiguous words by selecting the tag with the highest probability based on the surrounding context.

For large-scale chatbot deployment, HMM-based stochastic tagging is more suitable because it handles lexical ambiguity, learns from large datasets, adapts to different sentence structures, and provides more reliable tagging for real-world language.

4. **Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.**

Standardized POS tagsets provide a consistent way to identify grammatical categories such as nouns, verbs, adjectives, and adverbs. They help the chatbot understand sentence structure and identify the roles of important words. POS information improves intent detection by distinguishing actions from objects and supports better response generation. It also improves parsing, semantic analysis, and overall chatbot accuracy.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.
The sentence processed by the system is:
"Economic growth increases employment."
Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:
Change NNS to VBZ if the preceding word is tagged as NN.
The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

ANSWER:

1. **Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.**

Original POS tags:

economic/JJ
growth/NN
increases/NNS
employment/NN

Transformation rule:

Change NNS to VBZ if the preceding word is tagged as NN.

The word "increases" is tagged as NNS, and its preceding word "growth" is tagged as NN. Therefore, the transformation rule is applied.

Updated POS tags:

economic/JJ
growth/NN
increases/VBZ
employment/NN

The correction is linguistically appropriate because "increases" functions as a third-person singular verb describing the action performed by the subject "growth".

2. **Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.**

The initial error is the tagging of "increases" as NNS, which represents a plural noun. In the sentence "Economic growth increases employment", "economic" is an adjective modifying "growth", "growth" is the noun acting as the subject, "increases" is the verb expressing the action, and "employment" is the noun acting as the object.

Therefore, the correct structure is:

economic/JJ
growth/NN
increases/VBZ
employment/NN

The VBZ tag is appropriate because "increases" is a third-person singular present-tense verb.

3. **Assuming the corpus contains the words economic (120 occurrences), growth (450 occurrences), increases (210 occurrences), and employment (380 occurrences), compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.**

Total frequency:

120 + 450 + 210 + 380 = 1160

Frequency distribution:

economic = 120 / 1160 = 0.103 or 10.3%

growth = 450 / 1160 = 0.388 or 38.8%

increases = 210 / 1160 = 0.181 or 18.1%

employment = 380 / 1160 = 0.328 or 32.8%

Frequency information helps estimate the probability of words occurring in a corpus. Frequent words provide stronger statistical evidence for language models and POS taggers. The frequency of a word along with its surrounding context can help determine the most likely grammatical category, improving the accuracy of probabilistic POS tagging.

4. **Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.**

Transformation-Based Tagging starts with an initial POS tagging and applies learned correction rules based on contextual information. In this case, it changes "increases/NNS" to "increases/VBZ", correcting the grammatical error. This improves tagging accuracy because the system uses both the initial prediction and contextual patterns.

Entropy can be used to measure uncertainty in POS tag assignments. Before applying the rule, there may be uncertainty between NNS and VBZ for "increases". After applying the transformation rule, the system becomes more confident in assigning VBZ. Therefore, lower entropy after correction indicates reduced uncertainty and greater confidence in the POS tagging result.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________